

Beyond Perplexity: Entropy-Based Evaluation of Base Language Models

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Abstract

Perplexity once anchored language-model evaluation in predictive cross-entropy, but its token denominator prevents comparisons across tokenizers. Fixed task benchmarks provide common test sets, yet each observes only selected capabilities and can be distorted by benchmark-specific optimization or training-data overlap. We return base-model selection to the language-modeling objective with three adjustments: normalize cross-entropy by UTF-8 bytes, evaluate dynamically on fresh target-domain data before and after controlled adaptation, and filter contamination before scoring. Across 11 base models from 0.5B to 35B, adapted BPB tracks sentence-continuation quality closely ($r = -0.96$ against HellaSwag), while knowledge and arithmetic benchmarks reveal complementary strengths. Token-level analysis gives the reported quantities distinct roles: adapted BPB selects the model; zero-shot→adapted reduction describes corpus calibration. A fixed-target check shows that 512-token adapters retain their gains through 2,048-token inference (a 9.6% BPB reduction for Gemma-4-12B and 1.7% for Qwen-2.5-1.5B). Under one shared budget on news, Reddit, and Hacker News, adaptation materially reorders candidates within each corpus (zero-shot versus adapted Spearman $\rho = 0.609\text{--}0.682$), while the adapted ranking is highly stable across corpora ($\rho = +0.991, +0.955, \text{ and } +0.964$). The resulting rule is simple: rank models by adapted BPB on recent target-domain data and resolve near-ties within a tier using the target domain.

1 Introduction

Language models were originally judged by how well their predictive distributions compressed text: cross-entropy and its exponentiated form, perplexity. As evaluation shifted toward fixed task suites, scores became easier to communicate but narrower in scope. MMLU, HellaSwag, and GSM8K each expose a useful slice of capability rather than the full quality of a base language model [18, 20, 21]. Once a fixed benchmark also becomes a development target, benchmark-specific optimization and training-data overlap further weaken its value as an independent measure [15].

We argue for going back to the language-modeling objective, but not to raw perplexity. Modern base models span dense Transformers [30], sparse Mixture-of-Experts (MoE), Liquid Neural Networks, and incompatible tokenizers. A useful entropy-based comparison therefore needs three adjustments: a common unit, fresh and relevant evaluation data, and protection against memorized text.

Our framework supplies those adjustments. First, **Bits-Per-Byte (BPB)** normalizes cross-entropy onto UTF-8 bytes, so each model is measured against the same physical amount of text rather than its own tokenization. Second, a dynamic target-domain protocol replaces a permanently fixed test

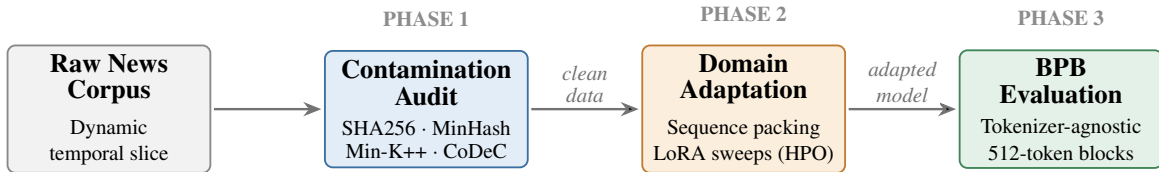


Figure 1: The Standardized Evaluation Pipeline Architecture. Contamination auditing (Phase 1) strictly precedes adaptation (Phase 2) and tokenizer-agnostic BPB evaluation (Phase 3).

with a recent corpus and records performance both before and after the same controlled LoRA adaptation. Third, contamination filtering removes suspected pre-training overlap before any sample can influence adaptation or scoring. The adapted BPB level is the primary selection score; the zero-shot→adapted reduction diagnoses how much corpus calibration the model needed.

Our contributions are: (1) an entropy-based evaluation protocol combining byte normalization, dynamic target-domain evaluation, controlled adaptation, and contamination filtering; (2) an 11-model comparison spanning 0.5B–35B and three architecture families; (3) token- and context-level analyses explaining calibration and transfer to 2,048-token inference; and (4) a three-corpus equal-budget validation showing that adaptation reorders candidates while adapted performance tiers transfer. Code, reproducible commands, and the public leaderboard are available at <https://github.com/DavidAfonsoValente/entropy-bench>.

2 Related Work

Entropy and task evaluation. Language modeling was originally grounded in byte- and character-level entropy [1], while token-level perplexity depends on each model’s vocabulary and segmentation [2, 4]. Normalizing log-likelihood by the UTF-8 byte restores comparability: The Pile popularized BPB as a cross-tokenizer metric [3], Paloma adopted it to measure LM fit [5], and byte-level architectures remove the tokenizer entirely [6, 7]. Community leaderboards measure complementary capabilities through MMLU [18], MMLU-Pro [19], HellaSwag [20], and GSM8K [21], often using the *lm-evaluation-harness* [17]. We treat these as complementary diagnostics rather than substitutes for the predictive objective. Our contribution is not BPB alone, but the protocol that makes entropy actionable for model selection: byte normalization, fresh target-domain data, controlled adaptation, a contamination gate, and cross-domain validation.

Contamination and adaptation. Large models verbatim-memorize pre-training text, increasingly so with scale and duplication [8, 9], making deduplication [10] and contamination auditing [15] prerequisites for honest evaluation. Our audit combines the reference-free detectors Min-K% and Min-K++ [11, 12] with behavioral completion- and quiz-based probes [13, 14]. Finally, domain-adaptive training reliably improves in-domain modeling [22], and scaling laws for transfer cast adaptation gains as a measure of a model’s generality [24]; we operationalize this cheaply and comparably across architectures via LoRA [23].

3 The Evaluation Pipeline

The pipeline (Figure 1) is a chronologically strict progression: no sample may influence adaptation or scoring before it clears the contamination gate.

3.1 Phase 1: Forensic Contamination Audit

We isolate data seen during pre-training with a hierarchical “Scout-then-Verify” framework:

- **Min-K++ Normalization (Scout):** We isolate tokens in the bottom $K = 20\%$ of probability and normalize them against the unconditional vocabulary distribution (μ, σ) at each step. High normalized likelihoods in historically rare tokens indicate verbatim memorization.
- **Contextual Confidence (CoDeC) (Scout):** We introduce CoDeC as an associative memory trigger. If a sequence S was memorized, providing its immediate context C causes a non-linear spike in predictive confidence (Δ_{CoDeC}).
- **Dynamic Choice Quiz (DCQ) (Gatekeeper):** Suspected samples are presented alongside synthetic distractors generated by adjacent word-swapping and random [PAD] insertion. If the model deterministically selects the original string, it is quarantined. The reported runs use two distractors per sample (a 1/3 chance level), the pipeline default.

The audit is run *per model*, and the *union* of all models’ verified leaks is quarantined globally, so every model is adapted and scored on one identical, forensically cleansed corpus. Because memorized samples are removed up front, the zero-shot→adapted BPB reduction reflects adaptation to the domain rather than recall of pre-training artifacts.

3.2 Phase 2: Controlled Domain Adaptation

The cleansed corpus is packed into contiguous $L=512$ -token blocks along EOS boundaries to maximize gradient density. For the news study, model-specific Optuna sweeps with **Successive Halving** explore LoRA rank $r \in \{8, 16, 32, 64\}$, α -multiplier $\in \{1, 2, 4\}$, dropout, learning rate (10^{-6} – 10^{-3} , log-scale), and effective batch size via gradient accumulation. Adapters are restricted to each model’s language backbone, and the selected configuration is trained under a cosine schedule (decay to $0.1 \times$ peak LR, gradient clip 1.0). The cross-domain confirmation in Section 7 uses one fixed configuration and the same 250-step budget for every model–corpus cell.

3.3 Phase 3: Tokenizer-Agnostic BPB Evaluation

Cross-entropy loss ($\mathcal{L}_{\text{nats}}$) aggregated over the held-out corpus is unified onto byte-space using the UTF-8 byte-length ($|B|$) of the source text:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln(2)} \times \frac{|\text{Tokens}|}{|B|} \quad (1)$$

Zero-shot and adapted BPB are both measured on the held-out test split, using a model-specific bytes-per-token factor estimated from the same test text, so scores are directly comparable within and across models. The packed test stream is evaluated as independent, non-overlapping 512-token blocks: losses are aggregated corpus-wide, while context resets at every block cut. Injected document-start markers remain available as context but are masked as prediction targets in the final cross-domain protocol. This makes the reported score an explicit 512-token-context comparison; Section 5.2 measures how performance changes within that window.

4 Experimental Setup

Models. We evaluate 11 base models spanning dense Transformers, a sparse MoE, and a Liquid architecture (0.5B–35B parameters); the full cohort is listed in Table 1.

Data. All models adapt to a single, semantically deduplicated slice of Google News dated 8 June 2026 (119,054 articles, ~ 361 MB of text), split 80/10/10 into train/validation/test (seed 42) with an enforced zero-intersection constraint between splits.

Static benchmarks. For the correlation study (Section 5.4) we score every base model with the *lm-evaluation-harness* (vLLM backend, bf16, greedy decoding, each tokenizer’s native BOS) on MMLU-Pro (5-shot CoT, a fixed 1,000-question subset, exact-match), HellaSwag (10-shot, length-normalized accuracy), and GSM8K (5-shot, flexible-extract).

Hardware. Adaptation ran under Slurm on the Leonardo (CINECA) GPU partition, 3 nodes $\times$ 4 NVIDIA A100 64 GB; the static-benchmark suite ran separately on A100 GPUs.

5 Results and Discussion

5.1 Adapted BPB Ranking

Table 1 and Figure 2 detail the performance across the 11-model cohort.

Model	Architecture	Params	Zero-Shot BPB	Adapted BPB	Rel. Improvement	Final Rank
Gemma-4-31B [27]	Dense Transformer	31B	0.753	0.583	22.6%	1
Gemma-4-12B	Dense Transformer	12B	0.867	0.618	28.7%	2
Ministral-3-14B [28]	Dense Transformer	14B	0.723	0.651	9.9%	3
Qwen-3.5-35B-MoE [29]	Mixture of Experts	35B	0.700	0.663	5.4%	4
Qwen-3.5-9B	Dense Transformer	9B	0.733	0.679	7.4%	5
Qwen-2.5-7B [26]	Dense Transformer	7B	0.753	0.728	3.3%	6
Qwen-3.5-4B	Dense Transformer	4B	0.775	0.757	2.3%	7
Llama-3.2-1B [25]	Dense Transformer	1.2B	0.814	0.773	5.1%	8
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.829	0.802	3.2%	9
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.346	0.854	36.5%	10
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.931	0.921	1.0%	11

Table 1: Model performance on the forensically cleansed news test set. Lower BPB indicates superior compression efficiency.

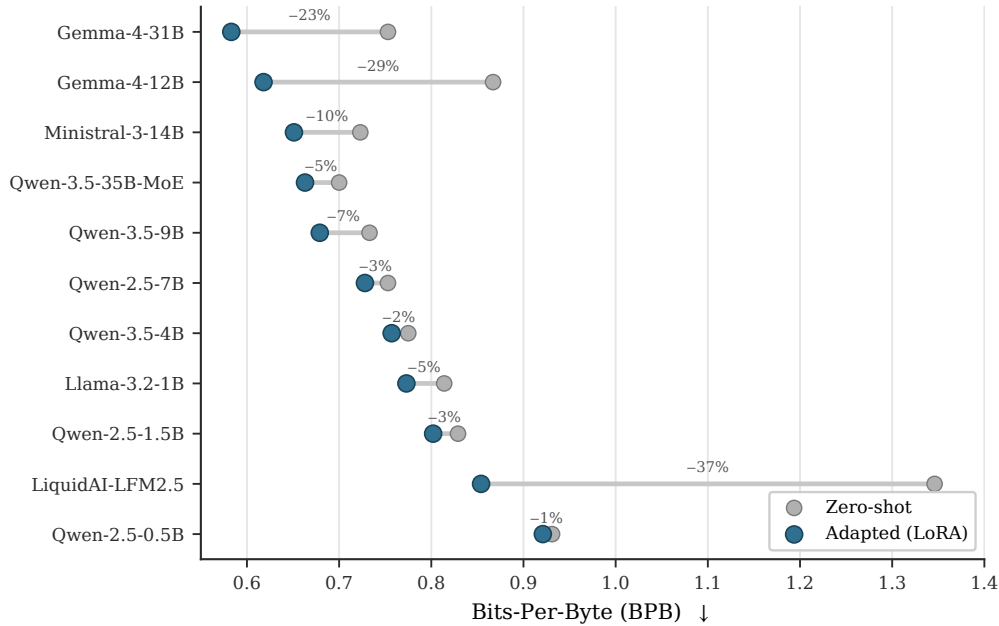


Figure 2: Zero-shot and adapted BPB for all 11 models. Lower adapted BPB is the selection criterion; the labeled reduction summarizes calibration to the news corpus.

The two reported values have separate jobs. **Adapted BPB** ranks the models we would deploy; **relative reduction** describes how much their distributions moved toward the corpus. Gemma-4-31B reaches the lowest adapted BPB (0.583), followed by Gemma-4-12B (0.618) and Ministral-3-14B (0.651). The sparse Qwen-3.5-35B-MoE places alongside dense 12–14B models, while LiquidAI-LFM2.5 shows the largest calibration shift (36.5%).

5.2 How to Read the Adaptation Reduction

To interpret the reduction, we score held-out blocks under each base model and its LoRA adapter, then decompose the per-token log-probability gain. The three analyzed models span the observed range and make the source of the movement visible.

Model	Loss (nats/token)		Red.	Share of the gain		
	base	adapted		marker	next 16	remainder
Gemma-4-12B	2.838	2.015	29.0%	3.6%	5.0%	91.3%
LiquidAI-LFM2.5	4.298	2.728	36.5%	1.1%	1.9%	97.0%
Qwen-2.5-1.5B	2.737	2.647	3.3%	0.0%	0.0%	100%

Table 2: Token-level decomposition of the adaptation gain over 4,000 test blocks. *marker* is an injected document-start token, *next 16* is the post-boundary window, and *remainder* is every other scored position. These document boundaries may occur anywhere inside a packed block.

Adaptation calibrates surface form. Gemma-4-12B’s largest movers are <bos>, punctuation, paragraph breaks, and common function words; LFM-2.5 shows the same pattern. Across the top 200 gaining tokens, formatting, punctuation, and function words account for 83% of the gain for both models, while content words account for 5.2% for Gemma and 14.0% for LFM. Qwen-2.5-1.5B begins closer to the corpus distribution and allocates more of its smaller gain to dates and figures. The reduction therefore gives a useful measure of corpus calibration. For Gemma, the marker and its next 16 tokens contribute $3.6\% + 5.0\% = 8.6\%$ of the measured gain. This is a post-document-boundary window, not the first 17 positions of every block; the marker itself is excluded from the final cross-domain BPB.

Context reveals steady-state quality. Figure 3 resolves loss and gain by position in the 512-token block and normalizes every bucket to UTF-8 bytes. Over the first eight positions, Gemma-4-12B averages 2.566 bits/byte and Qwen-2.5-1.5B averages 1.490; by positions 256–510, Gemma reaches 0.717 and Qwen 0.773. The crossing therefore remains after byte normalization and is not explained by vocabulary size alone. Once context is available, the 12B model shows its stronger steady-state language modelling. The opening 32 positions account for 23.0% of Gemma’s total gain and 12.9% of Qwen’s, making context recovery a clear component of the calibration signal.

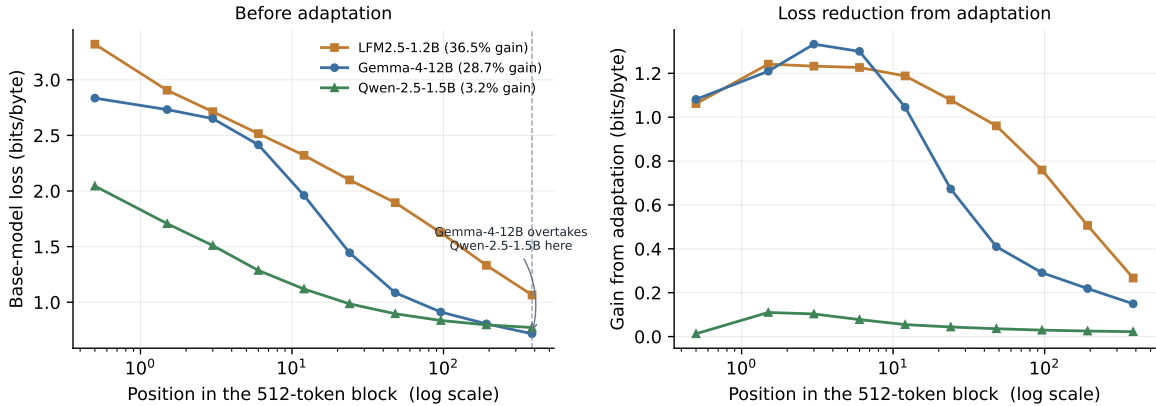


Figure 3: Base-model loss and adaptation gain by position in a packed 512-token block, measured in bits per UTF-8 byte. Positive gain is the average loss reduction from adaptation. The crossing persists after tokenizer normalization.

The 512-token adaptation transfers to longer inference contexts. We select 64 fixed held-out target windows and vary only the preceding context from 16 to 2,048 tokens, scoring the same target tokens at every length with exact byte normalization. Figure 4 shows that Gemma’s base BPB continues to recover beyond the training window, from 0.6465 at 512 tokens to 0.6054 at 2,048. Its adapted BPB is effectively unchanged ($0.5473 \rightarrow 0.5474$), so the adapter trained on 512-token blocks retains a 9.6% advantage at four times that context. Qwen’s smaller adaptation advantage also remains positive at 2,048 tokens ($0.7400 \rightarrow 0.7271$, a 1.7% reduction). Thus the base model continues to benefit from longer context, especially for Gemma, while the learned adaptation remains stable outside its training length.

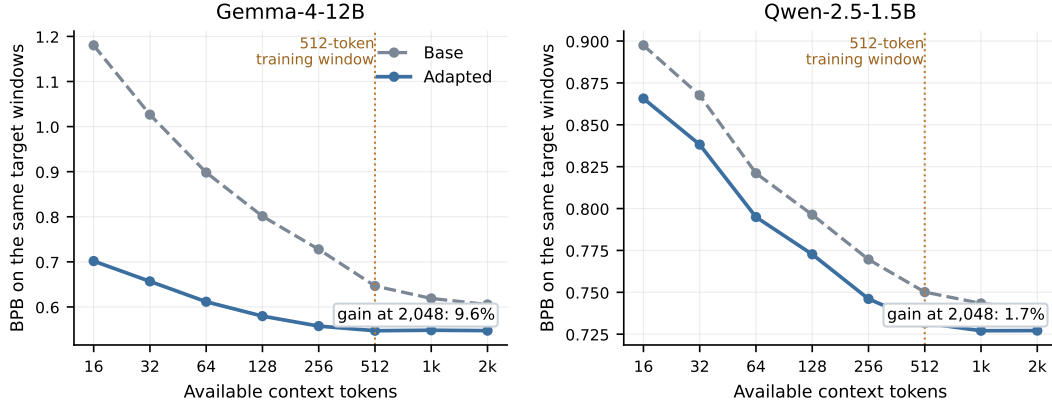


Figure 4: Fixed-target context-length check. Each point scores the same 64 held-out target windows; only the amount of preceding context changes. Both 512-trained adapters retain their improvement through 2,048 inference tokens.

Reporting rule. Rank models by the *adapted BPB level*. Use the zero-shot→adapted reduction as a calibration diagnostic that summarizes surface-form adjustment and context recovery. This two-number view makes both outputs easy to explain and apply.

5.3 Contamination Statistics: Verification Logic

Table 3 tracks forensic signals across all 11 models. We demonstrate that model size does not linearly correlate with memorization.

Model	Min-K++ Flags	CoDeC Flags	DCQ Candidates	DCQ Acc	Verified Leaks
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Mistral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

Table 3: Stage-by-Stage Forensic Breakdown, *test split* ($N = 1000$ sampled sequences per model per split; the audit itself runs on train, validation and test). Min-K++ and CoDeC “Scouts” flag anomalous candidates; DCQ acts as the final gatekeeper. Unique global leaks removed across all splits and models: 80 sequences.

5.4 Correlation with Static Benchmarks

We compare adapted BPB with **MMLU-Pro**, **HellaSwag**, and **GSM8K**. Table 4 uses the same nine-model subset for each rank comparison, while Figure 5 shows the full available scatter. The tasks agree most when they measure the same capability.

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K	Macro
Gemma-4-31B	0.583	0.612	0.865	0.817	0.764
Gemma-4-12B	0.618	0.583	0.840	0.778	0.733
Ministral-3-14B	0.651	0.541	0.829	0.807	0.726
Qwen-3.5-35B-MoE	0.663	0.620	0.855	0.865	0.780
Qwen-3.5-9B	0.679	0.562	0.816	0.860	0.746
Qwen-2.5-7B	0.728	0.475	0.803	0.813	0.697
Llama-3.2-1B	0.773	0.117	0.658	0.065	0.280
LiquidAI-LFM2.5	0.854	0.155	0.609	0.550	0.438
Qwen-2.5-0.5B	0.921	0.167	0.518	0.347	0.344

Table 4: Adapted BPB vs. standard benchmarks. Best per column in bold. Rank comparisons use the nine-model common subset reported across BPB and all three static tasks; Figure 5 shows the full available scatter.

Ranking the cohort by each task shows a smooth progression. Six of nine models keep the same rank under HellaSwag (sentence continuation), two under MMLU-Pro (knowledge), and none under GSM8K (arithmetic). The GSM8K shift is family-structured: every Qwen model moves up relative to its BPB position, while the Gemma, Mistral, and Llama models move down; Liquid rises by one place. This makes the metrics complementary. BPB measures general language modelling on the target corpus, while the task benchmarks add capability-specific information about knowledge and arithmetic.

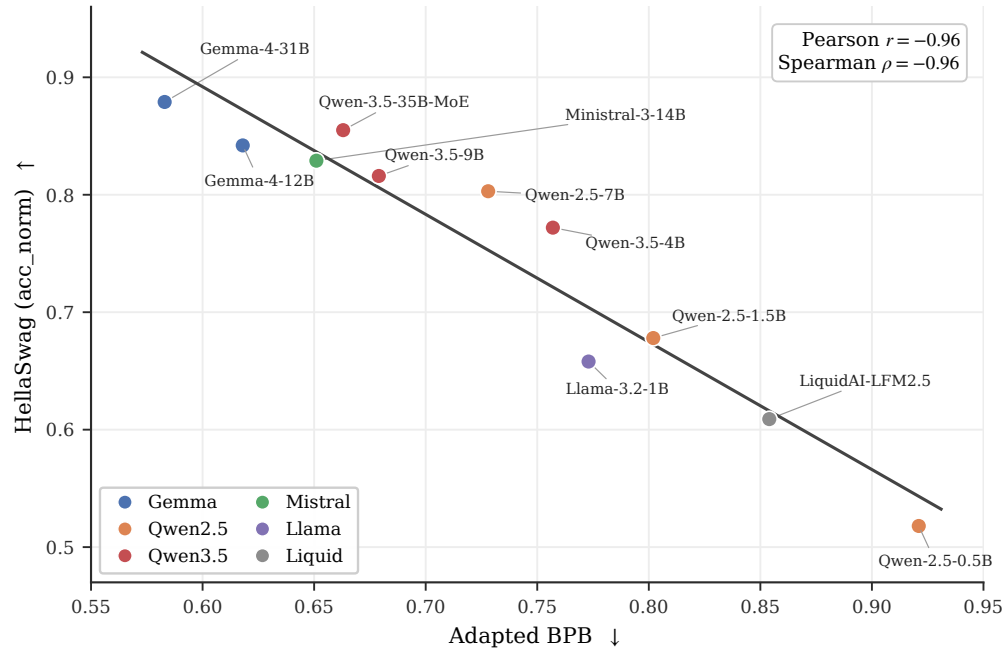


Figure 5: Adapted BPB vs. HellaSwag ($r = -0.96$). The strong association confirms that BPB tracks sentence-continuation quality.

6 A Reporting Standard for Entropy-Based Evaluation

The experiments support three conventions that make BPB comparisons easy to reproduce and explain.

Rank by adapted BPB. The adapted level measures the model after the same kind of domain intervention used in deployment. The reduction from zero-shot BPB remains useful as a separate calibration diagnostic.

Handle document boundaries consistently. Keep each model’s document-start marker as context, exclude the injected marker from the scored target, and normalize BPB over the same scored positions. The cross-domain study applies this convention in all 33 model–corpus cells.

Keep denominators fixed. Cross-domain ranks use all 11 models. Static-benchmark rank comparisons use the nine-model common subset reported across adapted BPB, MMLU-Pro, HellaSwag, and GSM8K, so the identical-rank counts (6, 2, 0) remain directly comparable.

7 Domain Transfer: Stable Ordering Across Corpora

Everything above rests on one domain. The claim under test is the one that would make BPB useful as a base-model selector rather than a news-specific instrument: *newer and larger models from well-resourced labs should outperform older and smaller ones in adapted BPB, and that ordering should be stable across domains*. We prepared two further corpora to the same grouping and cleaning recipe — exact deduplication on normalized text, Gopher quality heuristics, and MinHash/LSH near-duplicate filtering at a Jaccard threshold of 0.60 — then re-ran the cohort on all three.

Corpus	Raw records	Documents	Median words	Text
Google News, 8 Jun 2026	119,054	119,054	362	361 MB
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB

Table 5: Corpora for the domain-transfer study. Reddit and Hacker News comments are concatenated into document-length groups so all three corpora are evaluated at a comparable text scale. [†]The Reddit source contains four available days in the stated range.

Protocol. Every cell uses the same document-boundary convention, LoRA configuration (rank 16, $\alpha = 32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, all-linear targets), 250 training steps, and a 40,000-document cap. This fixed design makes the three-domain comparison direct: 11 models, three corpora, and 33 cells under one shared budget.

Adaptation changes the selection. Within each corpus, the zero-shot and adapted rankings are only moderately aligned: Spearman $\rho = 0.655$ on news, 0.609 on Reddit, and 0.682 on Hacker News, with maximum movements of seven, seven, and five places. Gemma-4-31B moves from fourth, fifth, and sixth zero-shot to first adapted on the three corpora; Gemma-4-12B moves from ninth on each zero-shot ranking to second, second, and fourth after adaptation. The adapted endpoint is therefore a materially different selection signal, not a rescaled zero-shot ranking.

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LiquidAI-LFM2.5	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 6: BPB across three domains under identical settings. Lower is better. All eleven models were adapted and evaluated on all three corpora.

The ordering is stable. Ranking all eleven models by adapted BPB within each corpus, news and Reddit agree on nine of eleven ranks (Spearman $\rho = +0.991$, maximum displacement one place). News against Hacker News gives $\rho = +0.955$ and Reddit against Hacker News $\rho = +0.964$, with a maximum displacement of two places. Gemma-4-31B ranks first on all three corpora.

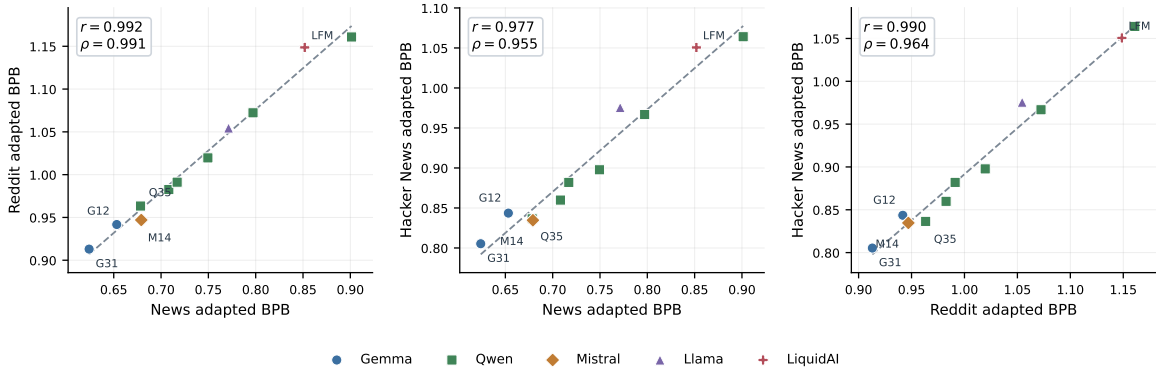


Figure 6: Pairwise adapted-BPB agreement across the three corpora. Pearson correlations are $r = 0.992$, 0.977 , and 0.990 ; rank correlations are $\rho = 0.991$, 0.955 , and 0.964 . Domain difficulty shifts the absolute values while the model ordering remains highly stable.

Every rank change occurs inside one of two near-tied clusters. The three mid-size models span 0.0259 BPB on news and 0.0089 on Hacker News; the two small models span 0.0258 and 0.0088. In both cases the spread narrows by roughly a factor of three. On Hacker News, Gemma-4-12B reaches 0.844, only 0.009 behind Ministral-3-14B and 0.008 behind Qwen-3.5-35B-MoE. This leads to a practical rule: adapted BPB identifies the top model and stable performance tiers, while a target-domain run resolves choices inside a near-tied tier.

The architecture results are easy to place. Below Gemma-4-31B, the adapted ranking follows model scale closely: the 12B, 14B, and 35B-MoE cluster, then 9B, 7B, 4B, the 1–1.5B group, and the 0.5B model. The sparse MoE scores alongside dense 12–14B models on every corpus, naturally locating its quality between its active and total parameter counts. LiquidAI-LFM2.5 shows a consistent 35–38% calibration gain across domains, illustrating why the adapted level and the reduction provide complementary information.

Domain difficulty and model order separate cleanly. All eleven models have their lowest BPB on news, highest on Reddit, and Hacker News between. Gemma-4-31B moves from 0.624 on news to 0.913 on Reddit, a 46% increase, while the largest rank displacement is only two places. Absolute BPB therefore describes domain difficulty, and relative order identifies the model to choose.

8 Conclusion

Entropy-based evaluation provides a broad, direct signal of base-model quality, but it becomes a fair model-selection tool only with the right protocol. Byte normalization makes cross-entropy comparable across tokenizers; fresh target-domain corpora keep the measurement relevant; contamination screening protects it from memorized text; and controlled adaptation measures the model in the state in which it will be used.

The 11-model study gives a clear interpretation of every output. Adapted BPB is the primary ranking; the reduction is a calibration diagnostic; the 512-token adaptation remains effective through 2,048-token inference contexts; HellaSwag confirms the language-modelling signal; and MMLU-Pro and GSM8K add capability-specific views. Under one shared budget across news, Reddit, and Hacker News, adaptation substantially changes the zero-shot candidate ordering, yet the resulting adapted ranking remains highly stable and Gemma-4-31B ranks first on every corpus.

The practical workflow is simple: collect recent target-domain text, clean and audit it, adapt every candidate under one budget, rank by adapted BPB, and resolve near-ties on the target domain. This produces a repeatable base-model decision from data the evaluator controls.

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